

Pranav Bhagwat

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SKILLS

Programming Languages: Python, C++, Java, Bash Shell, SQL, NoSQL (MongoDB), Dart, Latex
ML/SDE Frameworks: PyTorch, TensorFlow, Matplotlib, MMAction, OpenCV, HuggingFace, Wandb, ROS2
Systems: Distributed Systems, Microservices, REST APIs, OOP Design Patterns
Web/Other: Flutter, Flask, Generative AI, LLMs, Coding Agents
Cloud and DevOps: AWS (EC2, ECS, ECR), Docker, Apache Airflow, Git, GitHub Actions, CI/CD

WORK EXPERIENCE

Graduate Teaching Assistant - Generative AI Course | *North Carolina State University* **Jan 2026 - May 2026**

- Built and tested **end-to-end Python assignments** using PyTorch and HuggingFace, including starter scaffolding, reference solutions, and **automated test harnesses** to make grading deterministic and reproducible.
- Packaged **reproducible GPU environments** (dependency pinning, Colab/cluster setup) and debugged student issues spanning CUDA errors, dependency conflicts, and OOM failures.
- Wrote technical documentation and specs that translated research-level material into clear, incrementally testable engineering tasks.

Lead Software Engineer | *Thelios.ai (under Sportsseam LLC)* **Aug 2023 - Aug 2025**

- Architected and owned a **production Computer Vision microservice** on AWS (EC2, ECS, ECR) processing **100+ hours of footage weekly**, from model training through deployment and monitoring, achieving **95% accuracy in player detection** and **reducing manual analysis time by 80%**
- Engineered **temporal action localization models** using *CNNs, Vision Transformers and Vision-Language models in PyTorch*, achieving **85% accuracy** in classifying complex player behaviors.
- Implemented **custom graph-based heuristic algorithms** for team analysis.
- Expanded system capabilities across **multimodal understanding, detection, segmentation, tracking, pose estimation**, increasing coverage of player behavior analysis per play.

Software Development Intern | *BMC Software* **Jan 2022 - June 2022**

- Built a **Git-tracked end-to-end pipeline**: **NLP-based feature extraction** (NLTK), **ML anomaly classifier** (Scikit-learn), and iterative performance evaluation thereby achieving **86% accuracy on enterprise network config data**.
- Led a 4-intern team through full SDLC across a 4-month project timeline: requirements, implementation, testing, and iterative evaluation.

PROJECTS

OmniView: Instagram Event GraphRAG | *Neo4j, GraphRAG, Python, Streamlit, Groq, Whisper* **Jul 2026 - Present**

- Building an end-to-end **Graph Retrieval-Augmented Generation (GraphRAG)** pipeline that scrapes Instagram posts and Reels, transcribes audio via Whisper, and upserts structured entities into a **Neo4j Aura knowledge graph**.
- Implemented **hybrid semantic retrieval** combining vector similarity search on sentence-transformer embeddings with local graph traversal to surface conflicting claims and contradicting viewpoints across sources in a single query.
- Designed a **multi-perspective LLM synthesis layer** that explicitly highlights narrative gaps, opposing accounts, and source uncertainty, rendered in a **Streamlit chat UI** with inline citations linking every claim to its source post.

Sentinel AI - SolHacks 2026 Winner | *Python, TypeScript, Git, Gemini, Solana, FastAPI* **Mar 2026 - Mar 2026**

- Built a modular VS Code extension that intercepts GitHub Copilot agent's file actions post-inference, performing real-time contextual risk assessment against user-defined compliance policies before any code is executed - targeting regulated industries such as healthcare and fintech.
- Designed a **Python/FastAPI verifier backend** with a **hybrid policy engine (Regex + Gemini 2.5 Flash)** supporting hot-reloadable Markdown governance policies, **enforcing human-in-the-loop decision boundaries with deterministic, traceable risk verdicts**.
- Logged every governance decision immutably on-chain via Solana SPL Memo Program**; built a Chart.js analytics dashboard over on-chain data to give teams **full audit-trail visibility**.

EDUCATION

North Carolina State University **Aug 2025 - May 2027**
Masters in Computer Science (CGPA : 4.0) **USA**
Relevant Coursework: *Software Engineering, Operating Systems, Parallel Systems*

Pune Institute of Computer Technology **July 2019 - June 2023**
Bachelor of Engineering in Information Technology (CGPA : 9.31) **India**
Relevant Coursework: *DSA, OOP, Networks, OS, AI, ML, Computer Vision, Databases, Discrete Math*